

MCQ OF MEMBERSHIP OPERATOR :-

1. What is the output of the following code snippet?

```
Python
data = {"a": 1, "b": 2}
print(1 in data)
```

- A) True
- B) False
- C) TypeError
- D) None

2. Consider the following code. What will be printed?

```
Python
text = "Python Programming"
print("pro" in text)
```

- A) True
- B) False
- C) None
- D) Error

3. Which of the following statements is TRUE regarding the performance of the in operator?

- A) It is equally fast for lists and sets.
- B) It is faster for a list than a set because lists are ordered.
- C) It is significantly faster for a set than a list for large datasets.
- D) It cannot be used with a range object.

4. What is the result of this expression?

```
Python
nested_list = [1, [2, 3], 4]
print(2 in nested_list)
```

- A) True
- B) False
- C) TypeError
- D) [2, 3]

5. What is the output of the code below?

```
Python
x = "123"
y = 1
print(y in x)
```

- A) True
- B) False
- C) TypeError
- D) 123

6. Which of the following are the membership operators in Python?

- A) is and is not
- B) in and not in
- C) && and ||
- D) ++ and –

7. What will be the output of print('a' in 'apple')?

- A) True
- B) False
- C) None
- D) Error

8. What is the result of the following code?

```
Python
numbers = [10, 20, 30, 40]
print(50 not in numbers)
```

- A) True
- B) False
- C) None
- D) 50

9. Membership operators always return which data type?

- A) Integer
- B) String
- C) Boolean (True or False)
- D) Float

10. Fill in the blank to make the output True:

```
Python
colors = ["red", "blue", "green"]
print("red" _____ colors)
```

- A) is
- B) ==
- C) in
- D) inside

11. Which of the following are the identity operators in Python?

- A) in and not in
- B) == and !=
- C) is and is not
- D) and and or

12. What does the is operator actually compare?

- A) The values of two objects
- B) The memory addresses (ID) of two objects
- C) The data types of two objects
- D) The length of two objects

13. What is the output of the following code?

```
x = 5 y = 5
```

```
print(x is y)
```

- A) True
- B) False
- C) None
- D) Error

14. Which operator returns `True` if two variables do NOT point to the same object?

- A) is no
- B) not is
- C) !=
- D) is not

15. What is the output of this code?

```
list1 = [1, 2]  
list2 = [1, 2]  
print(list1 is list2)
```

- A) True
- B) False
- C) None
- D) [1, 2]

16. What is the output of the following code?

```
Python
```

```
a = [1, 2, 3]  
b = a  
c = a[:]  
print(a is b, a is c)
```

- A) True True
- B) True False
- C) False True
- D) False False

17. Consider this code snippet. What will be the output?

```
Python
```

```
x = "hello!"  
y = "hello!"  
print(x is y)
```

- A) True
- B) False
- C) None
- D) TypeError

18. Which of the following best describes the difference between == and is?

- A) == checks for identity; is checks for equality.
- B) == is for strings; is is for integers.
- C) == compares values; is compares memory addresses.
- D) There is no difference; they are interchangeable.

1. What is the output of the following code?

```
Python  
a = None  
print(a is None)
```

- A) False
- B) True
- C) None
- D) Error

20. How does Python evaluate the following?

```
Python  
print(type(True) is bool)
```

- A) False
- B) None
- C) True
- D) Error